



SUREN NETWORK
PLAY WITH TECHNOLOGIES

SNCNA

SUREN NETWORK CERTIFIED NETWORK ASSOCIATE

LAN COMMUNICATION TEST

1, Explain uses of ping command in network ?

- * ping command is major used to connectivity between two network devices, The protocol is used ^{ICMP}
- * It is used to check reachability, latency and ^{troubleshoot}

2, Explain netbios protocol ?

- * netbios → network Basic Input and output system
- * It is used ~~for~~ naming services to IP address in LAN → It is a legacy protocol. → works in port 137, 138, 139

3, Explain SMB protocol ?

- * server msg block → It is used for share files and printing in local network → no Internet needed

* Initial SMB server released then 1990's it runs separately over TCP/IP in port 445

4, Explain ping process from 50.0.0.10 to 50.0.0.20 ?

- 1) ping 50.0.0.20 → generate Echo → data or packet, which 32 bytes size → In Application layer of OSI layer model
- 2) In ~~session layer~~ → presentation layer → It defines what type or format of data.
- 3) session layer → Adds the frame to it and ~~network~~ ^{Transport} layer → It choose which type of communication should be made → (TCP or UDP) → connection or connectionless

Developed by SUREN NETWORK PVT LTD : mail@surennetwork@gmail.com
Whatsapp : 89 39 39 07 44 , 98 84 24 25 63

4) Network layer → Source and dest IP is added into data and called as packet → It is done Internet Protocol (IP)

5) Data link layer → Add the mac address of source and dest device → To find the mac ~~source~~ dest device → ethernet protocol which handle this whole process calls the Arp (Address resolution protocol) to find dest mac → Arp creates a request packet which has

Request

Source IP : 70.0.0.1

dest IP : 10.0.0.2

source mac : AOB

dest mac : FF:FF:FF → Broadcast mac

6) Arp request send as Broadcast, the devices which match IP of dest, will reply to Arp

request with reply packet of dest mac and add ~~mac~~ dest mac address into (ARP table)

7) ~~ethernet protocol~~

Arp table

8) ethernet protocol uses reply from arp with dest mac then send it to the physical layer

9) physical layer -> it converts binary or bits into electrical signals through physical cable or wireless and this data reaches target targeted IP address then OSI model process in reverse order in dest device.